

Datasheet TT-Motors with Encoder

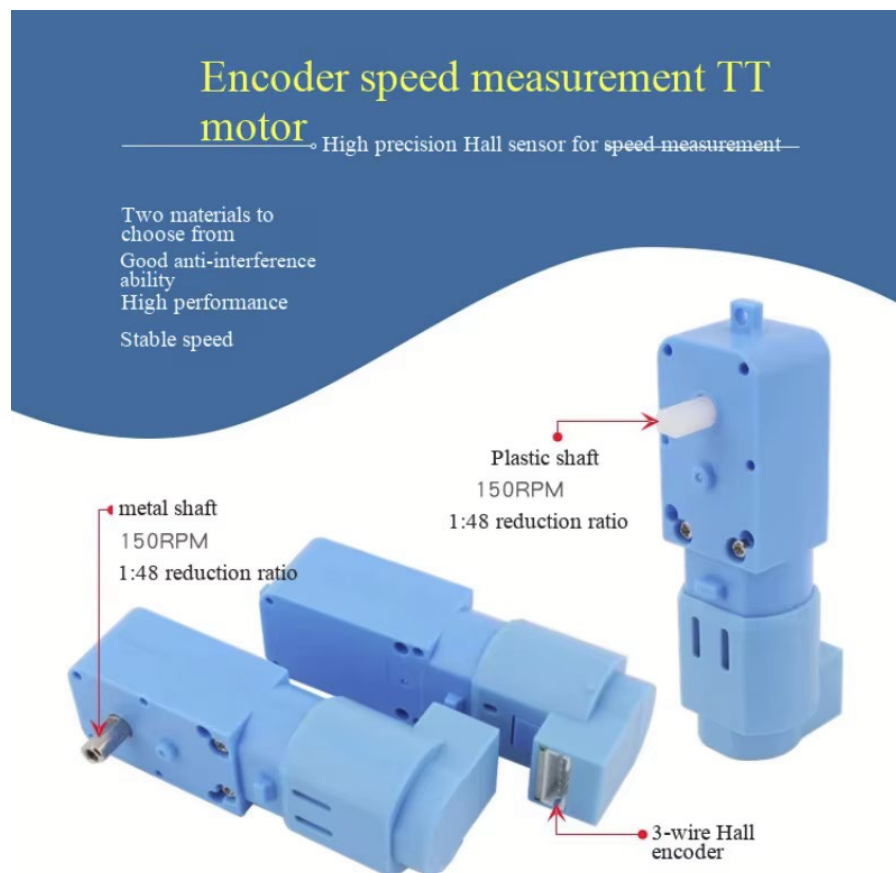
Manufacturer: n/a

Articel-Number: magnetic Encoder TT Motor

Link: [Magnetic Encoder TT Motor, Smart Car TT Motor, AB-Phase Incremental Hall Speed Measurement DC Gear Motor - AliExpress 502](#)



Abbildung 1: TT-Motor with Encoder

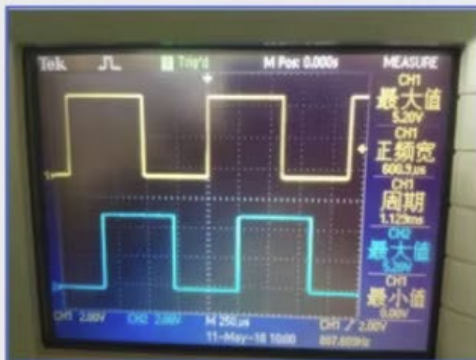


SPECIFICATIONS:

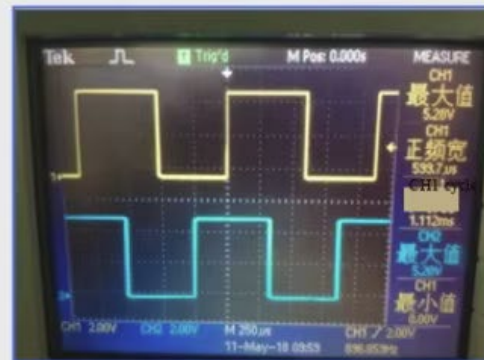
Product specifications and parameters

Products regulations
parameter

Plastic shaft  metal shaft 	
Motor name	3-wire Hall Speed TT Motor
Motor material	Plastic shaft/metal shaft(two materials to choose from)
reduction ratio	1:48 torque 1.2N. m
driving voltage	5V-12V(recommended 6-9V)
drive current	Rated 0.25A locked rotor current 0.6A
Comparison of rotational speed	Motor 7000±5%rpm, wheel axle 155±5%rpm
encoder type	Incremental Hall AB phase encoder
Encoder line count	3 lines
Featured Features	Equipped with pull-up shaping,the microcontroller can directly read it
Purchase recommendation	Plastic shaft:suitable for TT motor tires and TT motor specific Mecanum wheels Metal shaft:suitable for TT motor tires and Mecanum wheels with dedicated couplings

ENCODER:**Incremental Hall AB phase encoder****Magnetic Encoder Description**Magnetic encoder
instructions

Motor forward rotation



Motor reversal

If the phase difference between two signals is 90 degrees, then these two signals are orthogonal. The phase difference between the two signals is 90 degrees, so the direction of motor rotation can be determined based on the order of the two signals. The current tire travel distance can be calculated based on the number of signal pulses per unit time and the circumference of the tire. If only the number of AB phase pulses per unit time is detected, the current motor speed can also be measured.

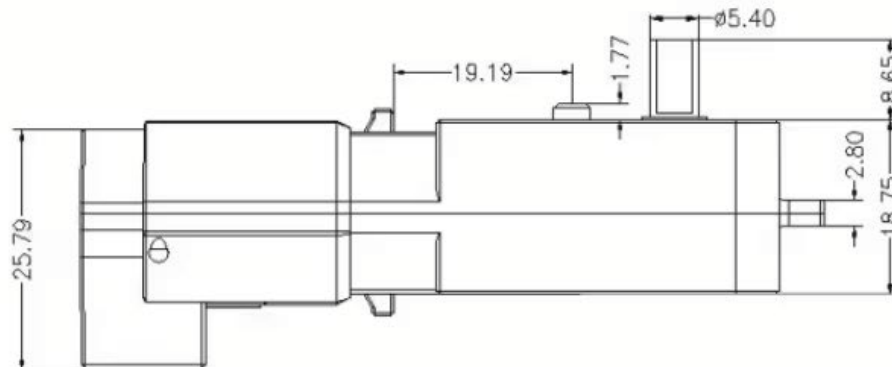
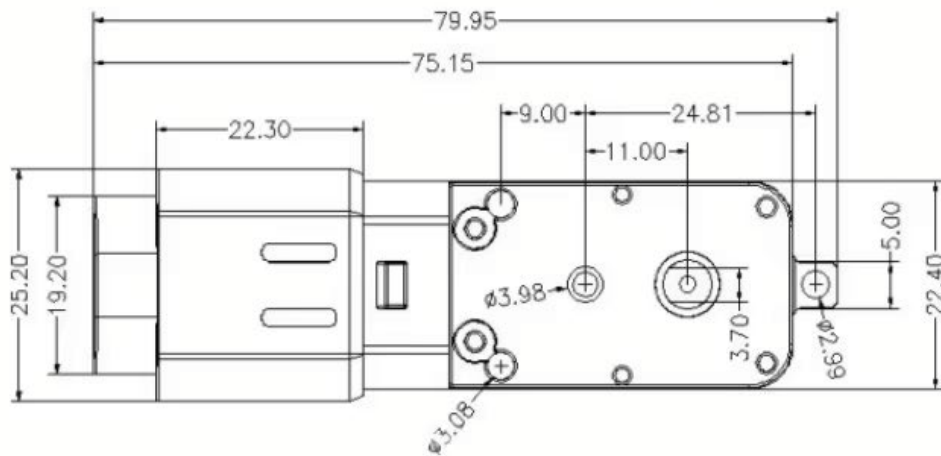
Taking the three wire encoder TT deceleration motor as an example, the motor outputs 3 pulses per single phase for one revolution, with a reduction ratio of 1:48. The maximum output of the motor output shaft for one revolution (reduction ratio * pulse number * multiplier) is $48 * 3 * 4 = 576$ pulses. The phase difference between the AB two-phase output pulse signals is 90° , which can detect the direction of motor rotation.

DIMENSIONS:

• Welding free standardized PH2.0 connection

Product size details

Product size
Details



PINOUT:

(five)

Welding free PH2.0 spacing connection

Motor interface description

Motor interface
Instructions



- ① Encoder ground wire
- ② Encoder output phase A
- ③ Encoder output B-phase
- ④ Positive pole of encoder
- ⑤ Motor wire+
- ⑥ Motor wire-

Wiring prompt:

M+ and M2 are connected to the motor driver output, A and B are connected to the microcontroller I/O port, VCC is connected to the encoder power supply 3.3-5V, and GND is the encoder power supply negative pole; If no encoder is needed, simply connect M+ and M - to drive.